

## Project Continuum

# Atoms to Inference

## MicroFi and the Physical AI Data Fabric

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### Core thesis

Physical AI succeeds only when the data that describes the physical world is trusted from the moment it is sensed.

MicroFi is the architectural proof point that the Cloudera data-in-motion family can extend below gateways and edge servers into the microcontroller-class sensing layer.

## EXECUTIVE NARRATIVE

# MicroFi completes the PADF story

The Physical AI Data Fabric should be described as Cloudera's answer to a simple but unforgiving truth: inference quality is bounded by the quality of the physical-world data that reaches the model. A camera frame, vibration sample, GPIO transition, CSI tensor, or temperature window is not yet AI-ready just because it has been captured. It becomes AI-ready when it carries identity, time, context, provenance, schema, retention policy, and a governed path into reasoning, replay, and retraining.

MicroFi is useful because it makes that statement concrete at the smallest practical power envelope. It extends the NiFi and MiNiFi data-in-motion family down into ESP32-class devices, where data is closest to the atoms. That does not mean Cloudera should expect broad customer installation of MicroFi on third-party industrial devices from incumbents such as Honeywell, Siemens, Rockwell, or Schneider Electric. The strategic value is different: MicroFi proves that Cloudera understands the entire Physical AI data chain, including the part most enterprise AI platforms ignore.

**Positioning sentence for the deck**

Cloudera does not just move data to models. Cloudera governs physical-world data from atoms to inference, across microcontrollers, edge gateways, on-prem ground stations, open storage, and retraining loops.

|                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>What MicroFi adds</b></p> <p>A name for the atomic edge of PADF.</p> <p>A technical proof that FlowFile-style data movement can begin below the gateway.</p> <p>A way to talk about metadata, provenance, durability, and governance at the sensing boundary.</p>          | <p><b>What MicroFi is not</b></p> <p>Not the primary enterprise adoption path for industrial customers.</p> <p>Not a replacement for incumbent OT devices or safety controllers.</p> <p>Not a promise that Cloudera will manage every sensor directly.</p> |
| <p><b>Why it matters</b></p> <p>Physical AI will fail if edge evidence is unaudited, unlabeled, unstructured, or stranded.</p> <p>The first mile is where trust is either created or lost.</p> <p>MicroFi turns "edge-aware" from a slogan into a demonstrable architecture.</p> | <p><b>How to say it</b></p> <p>NiFi governs enterprise flows.</p> <p>MiNiFi governs edge and gateway flows.</p> <p>MicroFi shows the same fabric can start at the atomic sensing layer.</p>                                                                |

# From Data in Motion to Physical AI Data Fabric

PADF is the governed operational-data loop stretched across every tier where physical-world intelligence is created.

## Data in motion from atoms to inference

MicroFi completes the atomic edge; MiNiFi and NiFi carry governed evidence into on-prem reasoning and retraining.



The existing PADF strategy names the loop as sense -> stream -> govern -> contextualize -> infer -> act -> audit -> retrain. MicroFi sharpens the first word in that loop. It asks what it means to sense in a way that is already metadata-rich, durable, policy-aware, and ready for downstream AI systems.

# Where each tier fits

The fabric is strongest when each tier does the work appropriate to its latency, power, sovereignty, and governance constraints.

| Tier                         | Representative assets                                                        | Data role                                                                                                                    | PADF message                                                              |
|------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Atomic sensing edge          | MicroFi on ESP32-class MCUs; CSI, GPIO, I2C, IMU, BLE, environmental sensors | Create FlowFiles or FlowFile-like evidence at the signal boundary; attach provenance, labels, schema, and retention posture. | The fabric begins before the gateway.                                     |
| Far edge / cell edge         | MiNiFi on Jetson, robot, camera gateway, industrial PC                       | Run local inference, filter high-volume streams, buffer during disconnects, route evidence selectively.                      | Move only the evidence that matters; keep normal data cheap.              |
| Private ground station       | Cloudera AI On-Prem, NiFi/MiNiFi gateway, NVIDIA NIM endpoints               | Perform multimodal reasoning, policy-aware adjudication, operator explanation, and bounded command dispatch.                 | On-prem reasoning is the natural middle tier for sovereign physical data. |
| Streaming and state backbone | NiFi, Kafka, Flink, operational stores                                       | Join live evidence with asset, mission, policy, and context state.                                                           | Inference needs context, not anonymous sensor payloads.                   |
| Governed corpus and replay   | SDX, Iceberg, object storage, model artifact stores                          | Preserve lineage from source signal to model output, operator action, and retraining example.                                | Audit and learning are data products, not afterthoughts.                  |

# MicroFi is an architectural proof point, not a volume adoption motion

Its value is to make the PADF claim credible at the physical sensing boundary.

The customer-facing PADF story should not imply that every industrial customer will flash MicroFi onto embedded devices they buy from established OT vendors. In most cases, customers will continue to run their existing controllers, cameras, sensors, gateways, and vendor-managed systems. PADF should meet those systems through MiNiFi, NiFi, protocol adapters, gateways, historians, message brokers, and partner integrations.

MicroFi serves a different strategic job. It shows that Cloudera understands the edge deeply enough to model the first mile from the bottom up. It gives the presentation a credible answer when the audience asks, "How far down does your data fabric really go?" The answer is: as far down as the point where atoms become digital evidence.

|                                                                                                                                                                                        |                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Completes the stack narrative</b><br>NiFi: enterprise and data-center flows.<br>MiNiFi: gateways, robots, cameras, edge servers.<br>MicroFi: microcontroller-class sensing edge. | <b>2. Moves metadata to the source</b><br>Attach identity, timestamps, calibration, schema, location, and firmware context before data becomes detached from its origin.                          |
| <b>3. Makes sovereignty practical</b><br>Keep sensitive data local, summarize at the edge, buffer through disconnects, and escalate only governed evidence.                            | <b>4. Shows edge-native empathy</b><br>A fabric that understands flash, PSRAM, queue durability, intermittent WiFi, and low-power sensors is more credible than one that starts at cloud ingress. |

### Narrative guardrail

Do not pitch MicroFi as the product customers must buy to use PADF. Pitch it as a proof point, developer toolkit, research substrate, and reference implementation that demonstrates Cloudera can reason about data quality from the smallest edge node to on-prem AI.

# What MicroFi is technically

A clean-room, microcontroller-class data ingestion agent participating in the Apache MiNiFi C2 ecosystem.

MicroFi is described in the repository README as an exploratory microcontroller-class data ingestion agent for the Apache MiNiFi C2 ecosystem, implemented clean-room against MiNiFi protocol contracts such as FlowFile semantics and C2 heartbeat/ack, with ESP32 as the first target. That framing matters: the point is not to port all of MiNiFi to a microcontroller. The point is to preserve the semantics that make data movement governable, observable, and lifecycle-managed.

| Current capability area                                                               | What it proves for PADF                                                                                      |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Core runtime types: FlowFile, Queue, Session, Processor, static Registry              | The smallest edge node can participate in a flow-graph mental model rather than emitting anonymous payloads. |
| Minimal flow engine and embedded processors such as GenerateFlowFile and LogAttribute | The runtime can host curated processors appropriate to constrained devices.                                  |
| WiFi station bring-up and HTTPS C2 heartbeat to a configurable endpoint               | The atomic edge can be visible to edge flow management and fleet monitoring.                                 |
| Flow-definition parsing for NiFi versioned-flow snapshots and MiNiFi Config Version 3 | Field behavior can be updated through familiar flow lifecycle concepts.                                      |
| C2 operation dispatch for DESCRIBE/manifest and UPDATE/configuration                  | The node can receive and apply managed configuration from the broader MiNiFi/EFM ecosystem.                  |
| LittleFS durable storage with watermarks and heartbeat storage metrics                | Offline durability and operational visibility begin before the gateway.                                      |

Note: The repository status is pre-alpha; the framing above treats MicroFi as a strategic and architectural proof point, not as a production support commitment.

# Why durability and constraints matter

Physical AI does not happen on a perfect network with infinite compute.

A microcontroller-class node has a different failure model than a cloud service or even an edge server. It may have limited flash, limited RAM, intermittent WiFi, strict power budgets, and no operator nearby. That is exactly why MicroFi is strategically useful: it forces the PADF story to account for the physics of the first mile.

| Hardware tier                                    | Minimum profile                                                                                        | PADF relevance                                                                                       |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Tier 1: registration and base flow               | Any ESP32 variant with WiFi and at least 4 MB flash; base GenerateFlowFile -> LogAttribute style flow. | Proves the agent can register, heartbeat, and run a minimal managed flow at microcontroller scale.   |
| Tier 2: LittleFS durability                      | ESP32 with WiFi and at least 4 MB flash; LittleFS queue, with larger capacity on ESP32-S3.             | Proves offline continuity: data can survive disconnects and power-cycle scenarios.                   |
| Tier 3: Physical AI Data Fabric P0 processor set | ESP32-S3 with 8 MB PSRAM and 16 MB flash; reference board uses 16 MB flash and 8 MB OPI PSRAM.         | Supports CSI windows, IMU buffers, camera frames, DSP, and lightweight on-device inference patterns. |

### Durability message

The MicroFi README describes a LittleFS-backed durable storage substrate with watermark eviction and heartbeat storage metrics, sized in the 16 MB layout for roughly 30 days of offline durability under typical edge-sensor rates. That is exactly the kind of first-mile resilience PADF should be known for.

This should be connected to the broader PADF story: resilience is not just a data center concern. A governed fabric is only as trustworthy as its weakest disconnected edge node. MicroFi shows how buffering, retention policy, flow identity, and operational metrics can exist at the sensing layer rather than being reconstructed later.

# From raw signal to AI-ready feature

The roadmap turns MicroFi from "small MiNiFi" into an atomic-edge Physical AI node.

The most important roadmap idea is that MicroFi-original processors do not exist merely because MiNiFi is too large. They exist because MiNiFi was not designed to live on a sensing endpoint. The MicroFi processor roadmap fills the gap between raw atoms and AI-ready evidence: CSI capture, sensor I/O, feature transforms, labels, provenance, distributed synchronization, egress, and observability.

| Processor family             | Example P0 processors                                                             | Why it matters                                                                                                                 |
|------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| CSI and WiFi-passive sensing | GetWiFiCSI, ExtractCSIAmplitudePhase, WindowCSI, DetectMotionCSI, RunBistaticPair | Turns commodity WiFi physics into a governed sensing modality for presence, motion, gesture, gait, and environmental features. |
| Sensor I/O                   | GetI2C, GetGPIO, GetIMU                                                           | Captures physical signals and ground-truth labels close to the source.                                                         |
| Feature and transform        | FFTContent, WindowAggregate, AttachLabel, AttachProvenance                        | Converts raw samples into windows, tensors, labels, and source-aware AI features.                                              |
| Distributed flow graph       | SiteToSiteLite, PublishFlowESPNow, SyncClockPTP                                   | Lets microcontroller nodes coordinate and preserve timing in multi-node sensing patterns.                                      |
| Egress and storage           | PutMQTTBatch, RotateLittleFS                                                      | Controls bandwidth, batching, replay, and offline durability.                                                                  |
| Health and observability     | GetESP32SystemMetrics, EmitHeartbeat                                              | Makes tiny devices manageable and diagnosable as part of a governed fleet.                                                     |

## The key shift

MicroFi is not about shrinking every enterprise connector into an ESP32. It is about building the processors that make a microcontroller useful to Physical AI: capture the signal, window it, label it, prove where it came from, survive disconnects, and emit it into the governed fabric.



# How MicroFi reinforces the existing reference architecture library

The current reference architectures do not need to be rewritten. MicroFi adds a lower sensing tier underneath them.

| Reference architecture                           | Existing PADF message                                                                                                                                | MicroFi extension                                                                                                                                                                                                                                               |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| From Edge Perception to Safe Shutdown            | MiNiFi selectively escalates anomaly evidence from a Jetson/cell edge to a private CAI On-Prem ground station for policy-gated safe action.          | MicroFi can add low-power local signals such as GPIO, fan vibration, enclosure temperature, current draw, proximity, door state, or CSI motion around the cell. The shutdown replay can then cite not only frames and OT tags, but atomic-edge sensor evidence. |
| Tactical Edge: Semantic Reasoning                | A rover uses MiNiFi to stream sampled frames and telemetry to a private ground station where Cosmos Reason and a mission agent plan bounded actions. | MicroFi nodes can serve as beacons, environmental sensors, IMU helpers, RF presence sensors, or peer sensing nodes that enrich the mission context without increasing raw video load.                                                                           |
| Industrial Near-Future Simulator                 | A live scene is converted into multiple short-horizon future rollouts, then scored for next-best-action guidance.                                    | MicroFi can provide non-visual context, such as temperature, vibration, proximity, occupancy, light, or RF motion, helping future rollouts become grounded in more than pixels.                                                                                 |
| Synthetic Training Data for Edge Smoke Detection | NiFi, Iceberg, SDX, CAI, Cosmos Transfer, and MiNiFi create, govern, deploy, and improve an edge smoke model.                                        | MicroFi can collect environmental labels and site context such as wind, light, humidity, particulate data, and device health, strengthening the labels and feedback loop used to retrain the model.                                                             |

The message to preserve is that the four reference architectures already prove PADF at the edge, on-prem, and governance layers. MicroFi makes those same stories stronger by showing that Cloudera can start the governed evidence chain below the camera gateway or robot computer when the scenario requires it.